Tertiary Infotech Academy Pte Ltd

Lab 01 — Set Up a Reproducible Finance Project

Python Programming for Finance · TGS-2026065049 · LO1 / A1

Scenario

A bank analytics team needs every learner to run the same package versions and verify the environment before analysing SGX data.

Goal

A verified Python project with an environment check and dependency record.

Tools

Python 3, uv or venv, Pandas, Matplotlib

Step-by-step

1. Create the lab project folder and open a terminal in it.

```
uv init python-finance-lab01
```

2. Add the analysis packages.

```
uv add pandas numpy matplotlib yfinance
```

3. Create env_check.py from the supplied starter file.

```
cp starter.py env_check.py
```

4. Run the environment check.

```
uv run python env_check.py
```

5. Record the Python and package versions in evidence.txt.

```
uv tree > evidence.txt
```

6. Review the output for missing packages or tracebacks.

Acceptance criteria

The script prints the Python, Pandas, NumPy and Matplotlib versions and ends with ENVIRONMENT READY.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.